```
import csv
from datetime import datetime

# CodeAlpha Internship - Task 2
# Stock Portfolio Tracker

STOCK_PRICES = {
    "AAPL": 180.00,
    "TSLA": 250.00,
    "GOOGL": 170.00,
    "MSFT": 420.00,
    "AMZN": 190.00,
    "NVDA": 140.00,
    "META": 560.00,
    "NFLX": 680.00,
    "ORCL": 175.00,
    "IBM": 215.00
}


def display_stocks():
    print("\nAvailable Stocks and Prices")
    print("-" * 40)
    print(f"{'Stock':<12}{'Price ($)':>15}")
    print("-" * 40)

    for stock, price in STOCK_PRICES.items():
        print(f"{stock:<12}{price:>15.2f}")

    print("-" * 40)


def get_positive_integer(message):
    while True:
        try:
            value = int(input(message))

            if value > 0:
                return value

            print("Please enter a quantity greater than zero.")

        except ValueError:
            print("Invalid input! Please enter a whole number.")


def calculate_portfolio():
    portfolio = []

    print("\nWelcome to Stock Portfolio Tracker!")
    print("Calculate the value of your stock investments.")

    display_stocks()

    while True:
        stock = input(
            "\nEnter stock symbol (or DONE to finish): "
        ).strip().upper()
```

```
        if stock == "DONE":
            break

        if stock not in STOCK_PRICES:
            print("Stock not found. Please select an available stock.")
            continue

        quantity = get_positive_integer(
            f"Enter quantity of {stock}: "
        )

        price = STOCK_PRICES[stock]
        investment = price * quantity

        portfolio.append({
            "stock": stock,
            "quantity": quantity,
            "price": price,
            "investment": investment
        })

        print(
            f"Added {quantity} shares of {stock}. "
            f"Value: ${investment:,.2f}"
        )

    return portfolio


def display_portfolio(portfolio):
    print("\n" + "=" * 65)
    print("                  STOCK PORTFOLIO REPORT")
    print("=" * 65)

    if not portfolio:
        print("No stocks were added to your portfolio.")
        return 0.0

    print(
        f"{'Stock':<10}{'Quantity':>10}"
        f"{'Price ($)':>15}{'Value ($)':>18}"
    )
    print("-" * 65)

    total_investment = 0.0

    for item in portfolio:
        print(
            f"{item['stock']:<10}"
            f"{item['quantity']:>10}"
            f"{item['price']:>15.2f}"
            f"{item['investment']:>18.2f}"
        )

        total_investment += item["investment"]

    print("-" * 65)
```

```
    print(f"{'TOTAL INVESTMENT:':<35}${total_investment:,.2f}")
    print("=" * 65)

    return total_investment


def save_portfolio_csv(portfolio, total):
    filename = "portfolio_report.csv"

    try:
        with open(filename, "w", newline="") as file:
            writer = csv.writer(file)

            writer.writerow([
                "Stock",
                "Quantity",
                "Price",
                "Investment Value"
            ])

            for item in portfolio:
                writer.writerow([
                    item["stock"],
                    item["quantity"],
                    item["price"],
                    item["investment"]
                ])

            writer.writerow([])
            writer.writerow(["Total Investment", "", "", total])

        print(f"\nPortfolio saved successfully to {filename}")

    except OSError as error:
        print(f"Could not save portfolio: {error}")


def main():
    while True:
        portfolio = calculate_portfolio()
        total = display_portfolio(portfolio)

        save_choice = input(
            "\nSave portfolio report to CSV? (yes/no): "
        ).strip().lower()

        if save_choice in ("yes", "y"):
            save_portfolio_csv(portfolio, total)

        again = input(
            "\nCalculate another portfolio? (yes/no): "
        ).strip().lower()

        if again not in ("yes", "y"):
            print("\nThank you for using Stock Portfolio Tracker!")
            print("Project: CodeAlpha Task 2")
            break
```

```
if __name__ == "__main__":
    main()
```